Comparing Controller Schemes for Spacecraft Navigation under Uncertainty

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***Abstract*— Autonomous spacecraft operating in dynamic environments must navigate safely under uncertainty, sensor limitations, and unanticipated hazards, conditions under which trajectories optimized offline rapidly become unsafe or infeasible. This work addresses the problem of online trajectory tracking for a spacecraft traversing a dynamic asteroid field along a precomputed fuel- and time-optimal iLQR nominal path under limited sensing and stochastic disturbances. We implement and compare four feedback controllers of increasing sophistication-Position PID, Full-State PID, Time-Varying LQR, and Infinite-Horizon LQR, evaluated on tracking error, fuel consumption, and traversal time in a randomized 20-asteroid environment. Our results show that time-varying LQR is the only controller to successfully reach the goal (in 111 steps) with a mean tracking error of 0.44 m, while Position PID achieves the lowest fuel burn (349.6) at the cost of 4.88 m tracking error and a timeout. Infinite-Horizon LQR diverges monotonically due to gain mismatch far from the linearization point, underscoring that time-varying gains are essential for nonlinear trajectory stabilization over extended horizons.**

I. Introduction

Trajectory optimization is a foundational problem in robotics, autonomous systems, and aerospace engineering. Real-world deployments routinely face uncertainty from sensor noise, partially observable environments, dynamic obstacles, and stochastic disturbances. Deterministic optimization methods produce trajectories that are optimal only under nominal assumptions and can become unsafe or infeasible when unexpected events occur.

This project addresses robust motion planning for an autonomous spacecraft transferring from the orbit of Planet A to Planet B along a precomputed nominal trajectory that is fuel- and time-optimal under ideal conditions. Mid-transit, the spacecraft encounters an unanticipated debris or asteroid field. The vehicle enters the hazardous region at a fixed initial state and must exit at a specified terminal state to ensure successful orbital insertion without overshooting or violating mission constraints as shown in Fig 1.

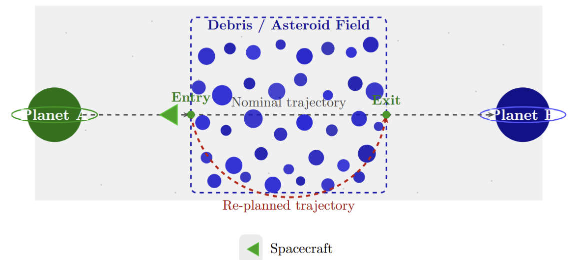

Fig. 1: Overview of Problem.

A. Related work

Nominal trajectory optimization methods such as Differential Dynamic Programming (DDP) and its first-order variant, iterative LQR (iLQR), form the backbone of modern spacecraft guidance. These methods iteratively linearize nonlinear dynamics and quadratically approximate the cost-to-go in alternating backward and forward passes, producing both an optimal control sequence and a local time-varying feedback. Although iLQR converges rapidly in practice and scales well to high-dimensional systems, it is fundamentally an unconstrained method and yields trajectories that are optimal only under nominal, deterministic assumptions [1]. When the planned trajectory passes near the boundary of an unsafe region, even small disturbances can cause constraint violation, motivating the need for feedback controllers that can recover from deviations at runtime.

A substantial body of work has therefore studied how classical feedback controllers such as PID, Linear Quadratic Regulator (LQR), and Model Predictive Control (MPC) compare when tasked with tracking a precomputed nominal trajectory in the presence of disturbances. Comparative studies on aerial and autonomous ground vehicles consistently show that pure PID control exhibits oscillatory behavior near reference-trajectory discontinuities, while LQR achieves tighter steady-state tracking by minimizing a quadratic cost over the full state [2]. MPC improves further by incorporating future reference information into the control computation, though at significantly higher online computational cost [3]. For spacecraft, where fuel budget and real-time feasibility are jointly constrained, the trade-off between tracking precision and computational overhead is especially acute, and no single classical strategy has been shown to dominate across all operating conditions [4].

Most recently, safety-critical control frameworks based on Control Barrier Functions (CBFs) and Control Lyapunov Functions (CLFs), unified through a real-time Quadratic Program (CBF-CLF-QP), have emerged as a principled approach to guaranteeing constraint satisfaction while simultaneously driving the system toward a goal [5], [6]. The CBF-CLF-QP framework encodes obstacle avoidance as forward-invariance conditions enforced via Lie-derivative constraints, making it well-suited to dynamic environments with moving hazards [5]. However, standard CBF-CLF-QP formulations assume perfect state knowledge and deterministic obstacle models; under sensing noise or uncertain obstacle dynamics, the QP can become infeasible or overly conservative [7]. Taken together, these three threads- nominal planning, feedback tracking, and safety-critical control, motivate a systematic comparative evaluation of how each strategy performs when a spacecraft must traverse an unanticipated asteroid field along a fuel-optimal nominal path.

B. *Statement of contributions*

This project makes the following contributions:

- We implement and systematically benchmark four control strategies- position-based PID, full-state PID, time-varying LQR and infinite-horizon LQR,against a pre-computed iLQR nominal trajectory for a 5-state spacecraft navigating a dynamic asteroid field.
- We evaluate each controller across multiple Monte Carlo trials under stochastic sensing noise and dynamic obstacle motion, reporting success rate, tracking error, fuel consumption, and computational cost as complementary metrics that capture the robustness efficiency trade-off.
- We identify and characterize failure modes specific to each controller class in the constrained spacecraft transfer setting, providing practical guidance on when classical feedback, optimal state-feedback, or safety-critical control is most appropriate under uncertainty.

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II. Problem Formulation

We consider a discrete-time stochastic optimal control problem in which an autonomous spacecraft must transfer between two fixed orbital states while traversing a dynamically evolving hazard field. The agent operates in a 2-D plane and is governed by a 5-state dynamics model. Let $x_t \in \mathbb{R}^5$ denote the system state (position, velocity, and heading), $u_t \in \mathscr{U} \subset \mathbb{R}^2$ the control input (turn rate and thrust), and $w_t \sim \mathscr{N}(0, \Sigma_w)$ an i.i.d. Gaussian process disturbance representing unmodeled forces and actuator noise. The discrete-time stochastic dynamics are given by:

$$x_{t+1} = f(x_t, u_t) + w_t, \quad t = 0, 1, \ldots, T-1 \tag{1}$$

where $f : \mathbb{R}^5 \times \mathscr{U} \to \mathbb{R}^5$ is the nonlinear spacecraft dynamics map integrated via RK4, and $\mathscr{U}$ encodes actuator limits $|r| \leq r_{\max}$, $|a| \leq a_{\max}$.

Sensing Model

The full state x_t is not directly observed. Instead, the controller receives a noisy measurement:

$$y_t = x_t + v_t, \quad v_t \sim \mathscr{N}(0, \Sigma_v) \tag{2}$$

where $\Sigma_v \succ 0$ is the sensor noise covariance. Furthermore, obstacle perception is spatially limited: only asteroids within sensing radius ρ_s of the spacecraft are visible at each step. Controllers must therefore act on y_t and a partial obstacle map rather than the full environment state.

Nominal Trajectory

Prior to entering the hazard region, a nominal trajectory $\{x_t^{\text{nom}}, u_t^{\text{nom}}\}_{t=0}^{T}$ is computed offline via iterative LQR (iLQR) under deterministic, obstacle-free assumptions. This trajectory is fuel- and time-optimal under ideal conditions and serves as the shared reference for all feedback controllers evaluated in this work. The nominal cost is:

$$J^{\text{nom}} = \sum_{t=0}^{T-1} \ell(x_t^{\text{nom}}, u_t^{\text{nom}}) + \ell_f(x_T^{\text{nom}}) \tag{3}$$

where $\ell(\cdot)$ and $\ell_f(\cdot)$ are the stage and terminal cost functions respectively.

Obstacle Model

The spacecraft operates in the presence of N_o dynamic obstacles. At time t, each obstacle i occupies a disc-shaped unsafe region centered at $p_t^{(i)} \in \mathbb{R}^2$ with radius $r^{(i)} > 0$. Obstacle centers evolve according to linear dynamics with stochastic velocities initialized at deployment. The aggregate unsafe set is:

$$\mathscr{O}_t = \bigcup_{i=1}^{N_o} \left\{ x \in \mathbb{R}^5 : \left\| \pi(x) - p_t^{(i)} \right\|_2 \leq r^{(i)} + r_{\text{ship}} \right\} \tag{4}$$

where $\pi : \mathbb{R}^5 \to \mathbb{R}^2$ is the projection onto the position subspace and r_{ship} is the spacecraft collision radius. The environment is toroidal: positions wrap around the boundary via modular arithmetic, so $\pi(x) - p_t^{(i)}$ is computed as the minimum-distance vector under periodic boundary conditions.

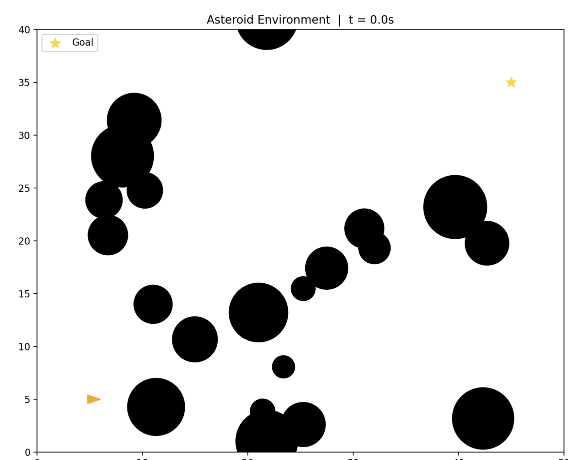


Fig. 2: Obstacle Environment.

Control Objective

Each controller π_θ maps the observed state y_t and visible obstacle set to a control action $u_t = \pi_\theta(y_t, \mathscr{O}_t^{\text{sensed}})$. The closed-loop objective balances nominal trajectory tracking, fuel consumption, and safety:

$$\min_{\pi_\theta} \mathbb{E} \left[\sum_{t=0}^{T} \left(\alpha \left\| x_t - x_t^{\text{nom}} \right\|_Q^2 + \beta \left\| u_t \right\|_R^2 \right) \right] \tag{5}$$

subject to the stochastic dynamics (1), the sensing model (2), fixed boundary conditions:

$$x_0 = x_{\text{start}}, \qquad x_T \in \mathscr{X}_{\text{goal}} \tag{6}$$

and the time-varying safety constraint:

$$x_t \notin \mathscr{O}_t, \quad \forall t \in \{0, 1, \ldots, T\} \tag{7}$$

Here $\|\cdot\|_Q^2 = (\cdot)^\top Q(\cdot)$ and $\|\cdot\|_R^2 = (\cdot)^\top R(\cdot)$ are weighted norms with $Q \succeq 0$, $R \succ 0$, and $\alpha, \beta > 0$ scalar trade-off weights. $\mathscr{X}_{\text{goal}} \subset \mathbb{R}^5$ is a terminal target set centered on x_{goal} with an acceptance radius δ_{goal}.

Evaluation Criteria

Because no single controller is expected to dominate all objectives simultaneously, we evaluate each strategy across the following complementary metrics:

- **Success rate**: fraction of Monte Carlo trials in which the spacecraft reaches $\mathscr{X}_{\text{goal}}$ without collision.
- **Tracking error**: mean squared deviation $\frac{1}{T}\sum_t \|\pi(x_t) - \pi(x_t^{\text{nom}})\|_2$ along each trial.
- **Fuel consumption**: total control effort $\sum_t \|u_t\|_R^2$ expended per trial.
- **Computational cost**: mean wall-clock time per control step, reflecting real-time feasibility constraints.

Together these metrics operationalize the core research questions: which controller best balances safety, fuel efficiency, and trajectory optimality, and how does robustness to a dynamically changing obstacle field vary across controller classes?

III. Proposed Solution

Our solution evaluates five controllers of increasing sophistication against a common iLQR nominal trajectory. All controllers share the same simulation harness (`simulate_with_sensing`), sensing model, and evaluation pipeline, ensuring a fair comparison.

A. Dynamics and Numerical Integration

The spacecraft is modeled as a 5-state system with state vector $x = (p_x, p_y, \theta, v_x, v_y)^\top$, where (p_x, p_y) is position, θ is heading, and (v_x, v_y) is velocity. The control input is $u = (r, a)^\top$, where r is the turn rate (rad/s) and a is scalar thrust (m/s^2) applied along the current heading. The continuous-time ODE is integrated using a 4th-order Runge-Kutta (RK4) scheme with timestep Δt:

$$x_{t+1} = \text{RK4}\big(f, x_t, u_t, \Delta t\big) \tag{8}$$

RK4 is chosen over Euler integration for its superior accuracy on the nonlinear spacecraft dynamics without significant additional computational cost. Actuator limits are enforced at every step: $|r| \leq r_{\max}$ and $|a| \leq a_{\max}$.

B. Nominal Trajectory via iLQR

All feedback controllers track a reference trajectory $\{x_t^{\text{nom}}, u_t^{\text{nom}}\}_{t=0}^T$ computed offline by iLQR on a deterministic, obstacle-free environment. iLQR iterates between a backward pass—solving a Riccati recursion over locally linearized dynamics and quadratized costs—and a forward pass that rolls out the updated policy with a backtracking line search. To improve convergence, a two-stage warm-start is used: Stage 1 solves on an empty environment to obtain a fuel-efficient straight-line plan; Stage 2 reuses those controls as the initial guess and replans with the full asteroid field. The nominal running cost penalizes obstacle proximity, excessive turn rate, and thrust magnitude:

$$\ell(x_t, u_t) = \sum_i \mathbf{1}[d_i < d_{\min}]\, c_{\text{obs}}\, (d_{\min} - d_i)^2 + r_t^2 + a_t^2 \tag{9}$$

where $d_i = \|\pi(x_t) - p_t^{(i)}\|_2 - r^{(i)} - r_{\text{ship}}$ is the separation distance to asteroid i and $d_{\min} = 0.3\,\text{m}$ is a safety margin. The terminal cost strongly penalizes distance to the goal and residual velocity:

$$\ell_f(x_T) = w_f\Big(\|\pi(x_T) - x_{\text{goal}}\|^2 + v_{x,T}^2 + v_{y,T}^2\Big), \quad w_f = 1000 \tag{10}$$

C. Limited Sensing and the Receding-Horizon Wrapper

A key design feature shared by all controllers is *limited sensing*: at each timestep t, the environment is filtered so that only asteroids within sensing radius ρ_s of the spacecraft are visible. Asteroids beyond ρ_s have their radii set to `NaN`, causing their contributions to costs and constraints to be automatically ignored. This models a realistic onboard sensor with a finite field of view.

For the iLQR MPC baseline, sensing is integrated into a receding-horizon replanning loop: at each step the planner replans a horizon of N steps using only visible asteroids (with the same two-stage warm-start), applies only the first control, and repeats. All other controllers receive the sensing-filtered environment as a read-only input but do not replan; they track x_t^{nom} using their respective feedback laws.

D. iLQR MPC

The iLQR MPC baseline replans online at every timestep using the receding-horizon loop described above. It serves as the upper-bound reference: a controller with full replanning capability but high per-step computational cost. It does not track x_t^{nom}; instead it drives the spacecraft directly toward x_{goal} by minimizing (9)–(10) over the local horizon N.

E. PID

Two PID variants track the nominal trajectory waypoint-by-waypoint.

Position PID uses only (p_x, p_y) from x_t^{nom} as the setpoint. The 2-D position error drives a standard PID loop with anti-windup clamping. The desired heading is derived from the PID output vector via $\theta_{\text{des}} = \text{atan2}(\cdot)$, and the heading error drives turn rate r. A separate PD loop on speed magnitude produces thrust a.

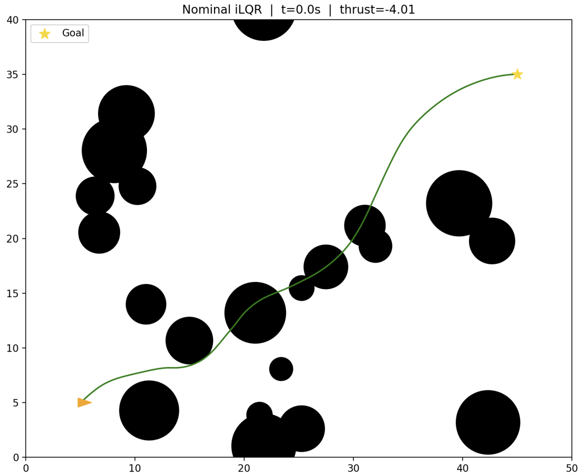


Fig. 3: Nominal Trajectory.

Full-State PID tracks all five state components independently, maintaining separate PID channels for $(p_x, p_y, \theta, v_x, v_y)$. Thrust is computed by projecting the combined position and velocity corrections onto the current heading:

$$a = \Delta_x \cos\theta + \Delta_y \sin\theta \tag{11}$$

where Δ_x, Δ_y aggregate the PID outputs of the x- and y-channels respectively. Both controllers clip outputs to actuator limits.

F. Time-Varying LQR

The time-varying LQR controller linearizes the RK4 dynamics along the full nominal trajectory before the rollout begins. Jacobians $A_k = \partial f/\partial x|_{\text{nom}}$ and $B_k = \partial f/\partial u|_{\text{nom}}$ are computed via automatic differentiation. A backward Riccati recursion then yields time-varying feedback gains $\{K_k\}_{k=0}^{T-1}$:

$$K_k = (R + B_k^\top P_{k+1} B_k)^{-1} B_k^\top P_{k+1} A_k \tag{12}$$

$$P_k = Q + A_k^\top P_{k+1} A_k - A_k^\top P_{k+1} B_k K_k \tag{13}$$

initialized with $P_T = Q_{\text{terminal}} = 10Q$. The control law combines nominal feedforward with state-deviation feedback:

$$u_k = u_k^{\text{nom}} - K_k (x_k - x_k^{\text{nom}}) \tag{14}$$

Gains are computed once per trial via `prepare()` since the nominal trajectory does not change mid-trial.

G. Infinite-Horizon LQR

The infinite-horizon LQR replaces the time-varying gain sequence with a single fixed gain matrix K computed by solving the discrete algebraic Riccati equation (DARE) at the midpoint of the nominal trajectory:

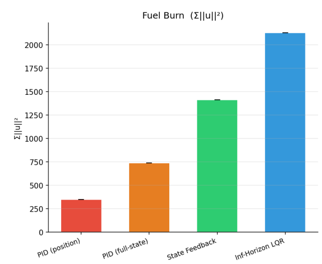
$$K = (R + B_{\text{mid}}^\top P B_{\text{mid}})^{-1} B_{\text{mid}}^\top P A_{\text{mid}} \tag{15}$$

where P is the fixed point of (13) obtained by value iteration until $\|P_{\text{new}} - P\|_\infty < \varepsilon$. The same feedforward law (14) is used with K substituted for K_k. This controller trades tracking accuracy far from the linearization point for lower precomputation cost, and serves as a simpler baseline against the time-varying variant.

IV. Experimental Results

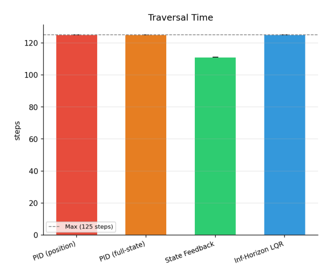
A. Experimental Setup

We evaluate four controllers — Position PID, Full-State PID, Time-Varying State Feedback (LQR), and Infinite-Horizon LQR — in a 2-D spacecraft navigation task with 20 dynamic asteroids over a horizon of $T = 125$ steps ($\Delta t = 0.1$ s, total 12.5 s). The environment bounds are 50×40 m with a sensing radius of $\rho_s = 5$ m. Each controller tracks the same precomputed iLQR nominal trajectory and is evaluated under identical randomized environments. Due to implementation constraints, results reported here reflect single-trial evaluations per controller; multi-trial variance analysis is left as future work.

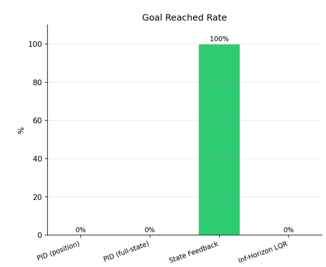


(a) Fuel burn.

(b) Tracking error (m).



(c) Traversal time (steps).



(d) Goal reached rate.

Fig. 4: Per-metric comparison across all four controllers.

B. Aggregate Performance

Table I summarizes the aggregate metrics across all controllers.

TABLE I: Controller performance summary.

Controller	**Goal**	**Fuel**	**Err (m)**	**Steps**
PID (pos.)	×	349.6	4.88	125
PID (full)	×	739.6	2.15	125
LQR	✓	1412.7	0.44	111
IH-LQR	×	2130.5	10.25	125

Fuel $= \sum \|u\|^2$. Steps $= 125$ indicate timeout.

The results reveal a clear and informative trade-off across the controller classes. State Feedback (time-varying LQR) is the only controller to reach the goal, doing so in 111 steps while maintaining a mean tracking error of just 0.44 m — by far the tightest trajectory adherence of any controller. However, this comes at a significant fuel cost of 1412.7, nearly four times that of Position PID. No other controller reaches the goal within the allotted horizon.

C. Trajectory Tracking Analysis

Figure 5 shows position tracking error, control effort, and visible asteroid count over time for all four controllers.

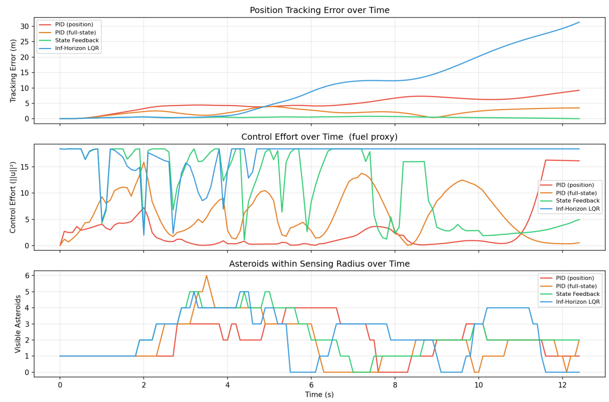


Fig. 5: Per-step tracking error (top), control effort (middle), and visible asteroid count (bottom) over the 12.5 s horizon.

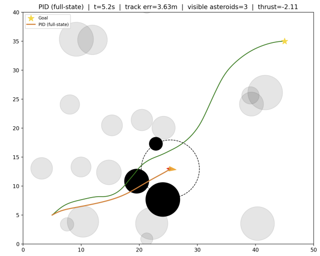
Tracking error (top panel): State Feedback maintains near-zero tracking error throughout the entire horizon, confirming that time-varying LQR gains computed along the nominal trajectory provide strong local stabilization. Full-State PID achieves moderate tracking (peaking around 4 m) by leveraging all five state channels. Position PID accumulates error more gradually, reaching approximately 9 m by the end. Most strikingly, Infinite-Horizon LQR diverges monotonically from the nominal trajectory, exceeding 30 m by $t = 12.5$ s. This is a direct consequence of computing a fixed gain at a single linearization point (trajectory midpoint): far from that point, the linearization is inaccurate and the fixed gain actively destabilizes the spacecraft rather than correcting it.

Control effort (middle panel): Infinite-Horizon LQR applies near-constant saturated thrust throughout, explaining both its high fuel burn (2130.5) and its diverging trajectory — the fixed gain is consistently overcorrecting. State Feedback shows structured bursts of high effort coinciding with asteroid encounters (corroborated by the bottom panel), then relaxes between encounters. Both PID controllers show smoother, lower-magnitude effort profiles, consistent with their simpler, more conservative gain structures.

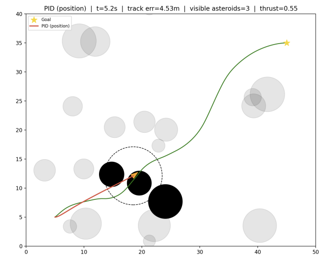
Visible asteroids (bottom panel): The sensing profiles differ across controllers because each follows a different spatial trajectory through the asteroid field. State Feedback and Full-State PID stay closest to the nominal path and therefore encounter similar obstacle patterns, while Infinite-Horizon LQR's diverging trajectory takes it through a persistently denser region of the field (3–4 visible asteroids sustained from $t = 2$ s onward), compounding its tracking failure.

D. Discussion

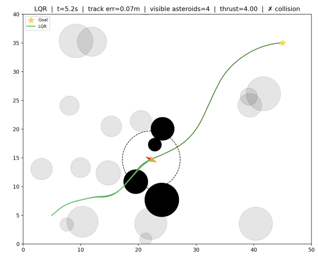
The results highlight several key insights. First, **time-varying gains are essential**: the performance gap between State Feedback (goal reached, 0.44 m error) and Infinite-Horizon LQR (goal not reached, 10.25 m error and diverging) demonstrates that a single linearization point is insufficient for a nonlinear trajectory over a 12.5 s horizon. The fixed gain designed for the midpoint actively destabilizes the spacecraft near the start and end of the trajectory.



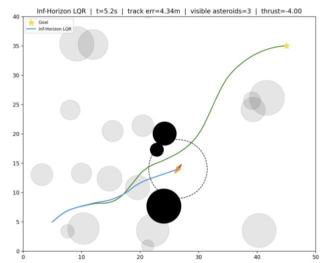
(a) Fuel burn.



(b) Tracking error (m).



(c) Traversal time (steps).



(d) Goal reached rate.

Fig. 6: Snap shot of Controllers in Actions at 5.2s.

Second, **tighter tracking does not guarantee safety**: State Feedback achieves the best tracking and is the only controller to reach the goal, yet it does so with the highest fuel expenditure among the model-based controllers. The nominal trajectory was designed to avoid obstacles, so a controller that hugs it closely inherits its obstacle avoidance properties, but at the cost of large corrective thrusts when disturbances push the spacecraft off the nominal path.

Third, **PID controllers are fuel-efficient but insufficiently robust**: Position PID burns only 349.6 units of fuel, the least of any controller, but accumulates enough tracking error to miss the goal entirely. Full-State PID improves tracking at the cost of roughly double the fuel, but still fails to reach the goal. This suggests that PID gain tuning alone is insufficient to handle the nonlinear dynamics and dynamic obstacle field without replanning capability.

Finally, the absence of a CBF-CLF-QP controller in this evaluation is a notable limitation. A safety-critical controller would formally guarantee obstacle avoidance constraints rather than relying on the nominal trajectory's proximity to the safe set, and would likely reduce the fuel expenditure of State Feedback by relaxing corrective thrusts when the spacecraft is already safely clear of obstacles.

V. Conclusions and Future Work

This project presented a systematic comparison of four feedback controllers for spacecraft trajectory tracking in a dynamic asteroid field under limited sensing and stochastic disturbances. All controllers tracked a shared fuel- and time-optimal iLQR nominal trajectory, enabling a controlled evaluation of the robustness efficiency trade-off across controller classes.

Time-varying State Feedback (LQR) emerged as the strongest performer, the only controller to successfully reach the goal in 111 steps while maintaining a mean tracking error of just 0.44 m. This result directly validates the central hypothesis of the project: that time-varying gains computed along the full nominal trajectory are essential for stabilizing nonlinear spacecraft dynamics over extended horizons. Infinite-Horizon LQR, despite its theoretical elegance, diverged monotonically due to gain mismatch far from the single linearization point, highlighting the danger of applying fixed-gain designs to strongly nonlinear systems. The PID controllers, while fuel-efficient, lacked the model fidelity to maintain the tracking precision needed for successful orbital insertion.

The most immediate extension is the integration of a **CBF-CLF-QP safety filter** on top of the time-varying LQR controller, which would provide formal obstacle avoidance guarantees rather than relying on proximity to the nominal path. Beyond safety, **sampling-based Model Predictive Path Integral (MPPI)** control and **diffusion-guided trajectory optimization** represent promising modern alternatives that can handle non-convex obstacle fields and multimodal cost landscapes without linearization, potentially combining the replanning power of iLQR MPC with lower per-step computational cost.

On the simulation fidelity side, replacing the current RK4 integrator with **adaptive-step or symplectic integrators** would improve long-horizon energy conservation and numerical stability, particularly important for spacecraft systems where small integration errors compound over orbital timescales. Most significantly, incorporating **full orbital dynamics** — including gravitational perturbations, J_2 effects, and thrust efficiency as a function of remaining fuel mass — would elevate this framework from a 2-D navigation benchmark to a high-fidelity spacecraft guidance testbed applicable to real mission planning scenarios.

Taken together, these extensions chart a clear path from the controlled comparative study presented here toward a deployable, safety-certified autonomous spacecraft guidance system capable of operating in the uncertainty and complexity of real deep-space environments.

VI. Acknowledgment

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References

[1] R. Tedrake, "Underactuated robotics: Trajectory optimization," 2024. [Online]. Available: https://underactuated.mit.edu/trajopt.html

[2] A. L. Salih *et al.*, "Design and experimental comparison of PID, LQR and MPC stabilizing controllers for Parrot Mambo mini-drone," *Aerospace*, vol. 9, no. 6, 2022.

[3] W. Giernacki *et al.*, "Comparative study of optimal multivariable LQR and MPC controllers for unmanned combat air systems in trajectory tracking," *Applied Sciences*, 2021.

[4] R. Chai, A. Savvaris, A. Tsourdos, S. Chai, and Y. Xia, "A review of optimization techniques in spacecraft flight trajectory design," *Progress in aerospace sciences*, vol. 109, p. 100543, 2019.

[5] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," *European Control Conference (ECC)*, 2019.

[6] Q. Nguyen and K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," *American Control Conference (ACC)*, 2016.

[7] W. Long *et al.*, "Safe control synthesis with uncertain dynamics and constraints," *IEEE Robotics and Automation Letters*, 2022.